

Quantum Computing Fundamentals

From kets to key distribution

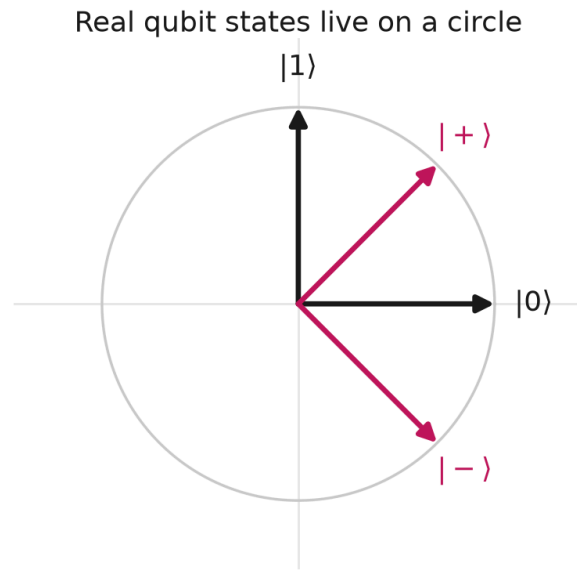
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Based on *Quantum Computing for Everyone* (Bernhardt), implemented in Qiskit

Everything here is 2D, and real

A qubit is a vector of length 1. With **real** amplitudes, every state sits on the unit circle. **Kets** are column vectors; **bras** are their transpose.

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \langle 0| = \begin{bmatrix} 1 & 0 \end{bmatrix}$$



Complex amplitudes would fill a sphere instead. Not needed until the quantum Fourier transform.

A qubit, and why the squares sum to 1

$$|v\rangle = c_0|0\rangle + c_1|1\rangle \quad c_0^2 + c_1^2 = 1$$

c_0 and c_1 are **probability amplitudes**. Measuring gives

$$P(0) = c_0^2 \quad P(1) = c_1^2$$

so the normalisation condition is just "the probabilities add to 1".

Standard basis — $\{|0\rangle, |1\rangle\}$ — is orthonormal:

$$\langle 0|0\rangle = 1 \quad \langle 1|1\rangle = 1 \quad \langle 0|1\rangle = 0$$

A bra times a ket gives a number: the overlap between two states.

Two qubits: the tensor product

Alice has $|v\rangle = c_0|a_0\rangle + c_1|a_1\rangle$, Bob has $|w\rangle = d_0|b_0\rangle + d_1|b_1\rangle$.

$$|v\rangle \otimes |w\rangle = c_0d_0|a_0b_0\rangle + c_0d_1|a_0b_1\rangle + c_1d_0|a_1b_0\rangle + c_1d_1|a_1b_1\rangle$$

Name the four amplitudes:

$$r = c_0d_0 \quad s = c_0d_1 \quad t = c_1d_0 \quad u = c_1d_1$$

It is still normalised, because

$$\underbrace{(c_0^2 + c_1^2)}_1 \underbrace{(d_0^2 + d_1^2)}_1 = r^2 + s^2 + t^2 + u^2 = 1$$

The entanglement test

If the state really came from a tensor product, then

$$ru = c_0d_0 \cdot c_1d_1 = c_0c_1d_0d_1 \quad st = c_0d_1 \cdot c_1d_0 = c_0c_1d_0d_1$$

They must be equal. So:

$$\begin{array}{l} ru = st \Rightarrow \text{separable} \\ ru \neq st \Rightarrow \text{entangled} \end{array}$$

Separable means each qubit has a state of its own. Entangled means only the **pair** has a state.

Equivalently: $ru - st$ is the determinant of $\begin{bmatrix} r & s \\ t & u \end{bmatrix}$, so separable = determinant zero.

Example A — not entangled

Alice $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, Bob $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. Tensor them:

$$r = \frac{1}{2\sqrt{2}} \quad s = \frac{\sqrt{3}}{2\sqrt{2}} \quad t = \frac{1}{2\sqrt{2}} \quad u = \frac{\sqrt{3}}{2\sqrt{2}}$$

Test: $ru = \frac{\sqrt{3}}{8}$ and $st = \frac{\sqrt{3}}{8}$. Equal $\rightarrow$ separable.

Group by Alice's kets:

$$|a_0\rangle \left(\frac{1}{2\sqrt{2}} |b_0\rangle + \frac{\sqrt{3}}{2\sqrt{2}} |b_1\rangle \right) + |a_1\rangle \left(\frac{1}{2\sqrt{2}} |b_0\rangle + \frac{\sqrt{3}}{2\sqrt{2}} |b_1\rangle \right)$$

Normalise each bracket — both have norm $\sqrt{\frac{1}{8} + \frac{3}{8}} = \frac{1}{\sqrt{2}}$ — and Bob's part is a **common factor**:

$$= \left(\frac{1}{\sqrt{2}} |a_0\rangle + \frac{1}{\sqrt{2}} |a_1\rangle \right) \left(\frac{1}{2} |b_0\rangle + \frac{\sqrt{3}}{2} |b_1\rangle \right)$$

Example B — entangled

$$r = \frac{1}{2} \quad s = \frac{1}{2} \quad t = \frac{1}{\sqrt{2}} \quad u = 0$$

Test: $ru = 0$ but $st = \frac{1}{2\sqrt{2}}$. Different $\rightarrow$ entangled.

Group by Alice's kets and normalise the same way:

$$\frac{1}{\sqrt{2}}|a_0\rangle \left(\frac{1}{\sqrt{2}}|b_0\rangle + \frac{1}{\sqrt{2}}|b_1\rangle \right) + \frac{1}{\sqrt{2}}|a_1\rangle|b_0\rangle$$

Bob's two brackets are now **different**, so nothing factors out. Neither qubit has a state of its own.

What that means: Alice measures $|a_0\rangle$ or $|a_1\rangle$ with equal odds — but her result decides Bob's. Get $|a_1\rangle$ and Bob is $|b_0\rangle$ with certainty.

Gates

So far: how to **describe** a two-qubit state, and test whether it is entangled.

Now: how to **change** one — and how entanglement actually gets made.



I and Z — leave the magnitudes alone

All of these act on $a_0|0\rangle + a_1|1\rangle$ in the standard basis.

I is the **identity** — it does nothing:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I(a_0|0\rangle + a_1|1\rangle) = a_0|0\rangle + a_1|1\rangle$$

Z leaves both **magnitudes** unchanged but flips the sign of a_1 — the **relative phase**:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad Z(a_0|0\rangle + a_1|1\rangle) = a_0|0\rangle - a_1|1\rangle$$

Probabilities are unaffected, since $|-a_1|^2 = |a_1|^2$. The phase only becomes visible once the qubit **interferes** with another — for instance after a Hadamard.

X and Y — the two NOTs

Both exchange $|0\rangle$ and $|1\rangle$.

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad Y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$X(a_0|0\rangle + a_1|1\rangle) = a_1|0\rangle + a_0|1\rangle \qquad Y(a_0|0\rangle + a_1|1\rangle) = a_1|0\rangle - a_0|1\rangle$$

So X is a pure swap; Y swaps **and** changes the relative phase.

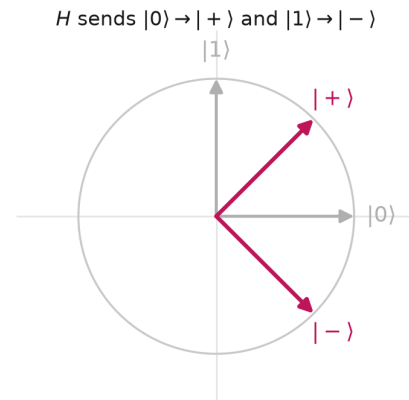
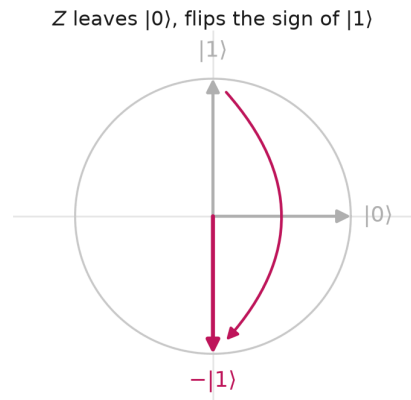
The Pauli Y is usually $-i$ times the matrix above, giving $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$. The real version is used here because these slides avoid complex numbers — but only the complex one squares to I . Qiskit's `qc.y()` applies the complex one.

Hadamard — into superposition

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$



In a diagram, a one-qubit gate is a square with its letter in it:



Why Hadamard, specifically?

The $\frac{1}{\sqrt{2}}$ is not a choice. Without it $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} |0\rangle$ has $\text{length}^2 = 2$, so the probabilities would sum to 2. Every gate must preserve length.

Plenty of gates make superposition:

	$P(0)$	$P(1)$
$R_y(\pi/3) 0\rangle$	0.75	0.25
$R_y(\pi/2) 0\rangle$	0.50	0.50
$R_y(2\pi/3) 0\rangle$	0.25	0.75

$R_y(\pi/2)$ gives the **same** equal superposition as H does from $|0\rangle$.

But only H undoes itself:

$$H^2 = I \quad R_y(\pi/2)^2 = R_y(\pi) \neq I$$

They agree on $|0\rangle$ and differ on $|1\rangle$ — that difference *is* the involution.

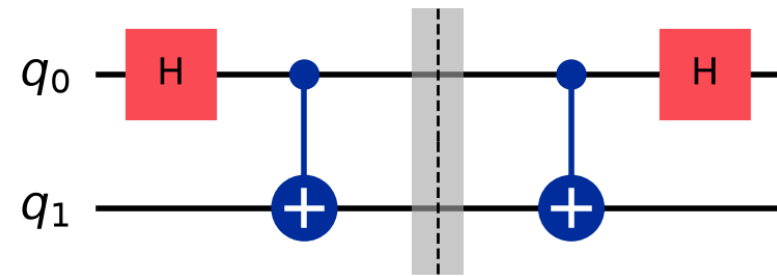
H is a **basis change** between Z and X , and swapping bases twice returns you home.

Every protocol here leans on that: the Bell circuit, Bob's decoder, BB84's X -basis measurement.

Involutions — gates that undo themselves

Applying one twice returns the input: $U^2 = I$

Gate	Qubits	What it does
X	1	swaps $ 0\rangle \leftrightarrow 1\rangle$
Y	1	swaps, with phases
Z	1	sign flip on $ 1\rangle$
H	1	basis change
CNOT	2	flip if control is 1
SWAP	2	exchange the two
Toffoli	3	flip if both controls 1
Fredkin	3	swap if control is 1

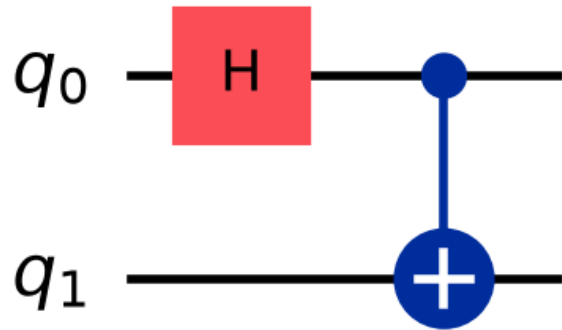


At the barrier the state is $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$.
Run the same two gates again and you are back at $|00\rangle$.

That "un-preparing" step is exactly what Bob does in superdense coding, and Alice in teleportation.

CNOT, and making entanglement

A single-qubit gate can **never** entangle. You need a two-qubit gate.

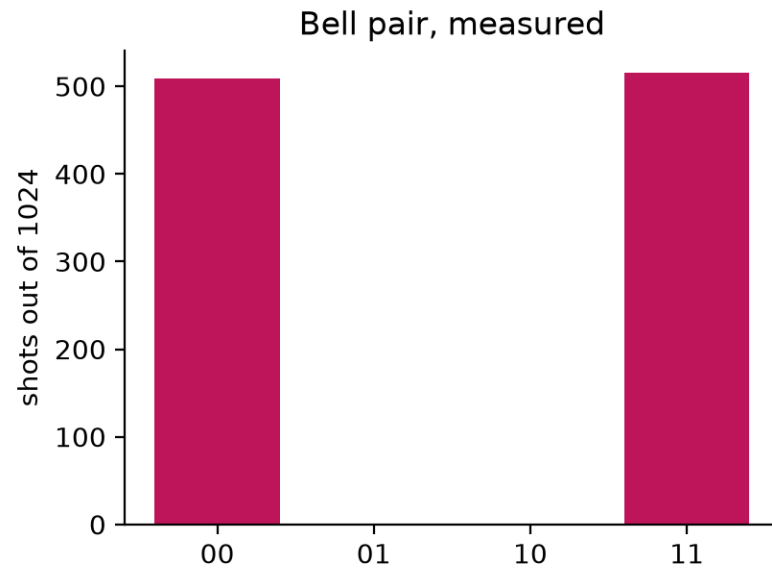


```
qc = QuantumCircuit(2)
qc.h(0)          # superposition
qc.cx(0, 1)      # entangle
```

$$|00\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Test it: $r = u = \frac{1}{\sqrt{2}}$, $s = t = 0$, so $ru = \frac{1}{2} \neq 0 = st$. **Entangled.**

And it shows up in the measurements



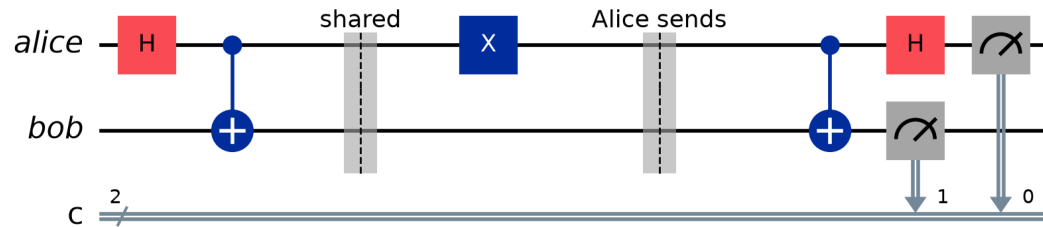
Only `00` and `11` . Never `01` or `10` .

The two qubits always agree — but neither had a value before measurement.

This **Bell pair** is the resource every protocol below is built on.

Superdense coding

Send **2 classical bits** by physically transmitting **1 qubit**.



Alice owns q0, Bob owns q1 — they share $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, one each.

Alice applies one gate to **her qubit only**, sends it to Bob, and he — now holding both — undoes the Bell circuit.

Bob finally reads	00	01	10	11
so Alice applies	I	X	Z	ZX

The top row is the **message Bob reads after decoding** — not the state in flight.

Superdense coding — why it works

$$|00\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice applies X to send `01`:

$$\xrightarrow{X} \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$$

Bob reverses the Bell circuit:

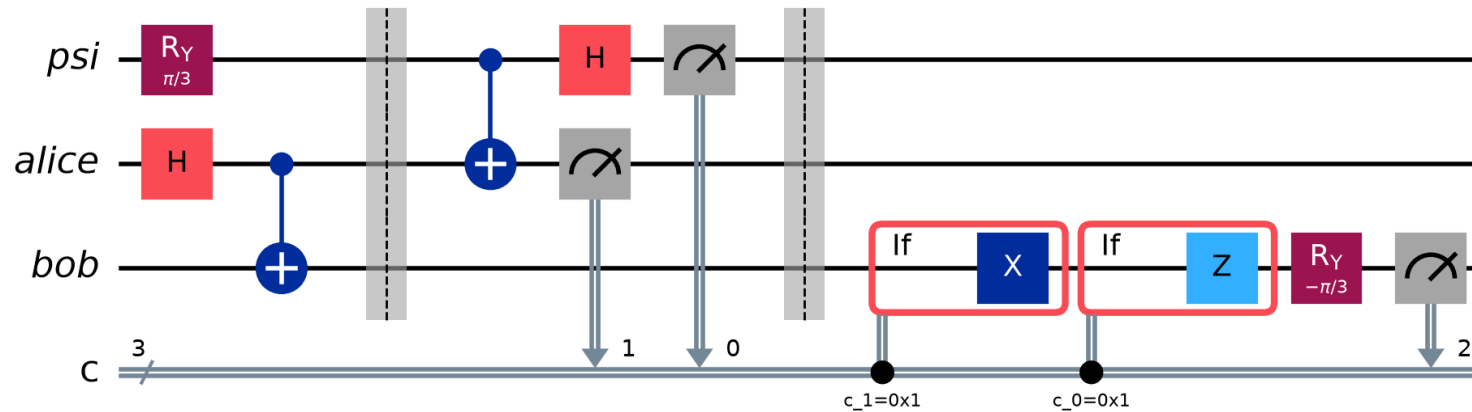
$$\xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|11\rangle + |01\rangle) \xrightarrow{H} |01\rangle \quad \text{— the two bits he reads}$$

```
qc.h(0); qc.cx(0, 1)    # Bell pair, shared in advance
qc.x(0)                 # Alice encodes "01" on her qubit alone
qc.cx(0, 1); qc.h(0)    # Bob decodes
```

The four gates map the pair onto four distinguishable Bell states. Entanglement is what lets one qubit carry two bits.

Quantum teleportation

Move **1 qubit's state** using **2 classical bits**. The exact opposite trade.



Alice owns *psi* and *alice* ; Bob owns *bob* . She entangles her state with her half of the pair, measures both of hers, and sends two classical bits — nothing quantum crosses. Bob corrects his qubit.

Bob receives	00	01	10	11
Bob applies	I	X	Z	Y

Teleportation — the catch

The state is **not copied** — no-cloning forbids that. Alice's measurement destroys her copy at the moment Bob's appears.

Nothing travels faster than light either: without the 2 classical bits, Bob's qubit is useless. Each of his four possible states is equally likely.

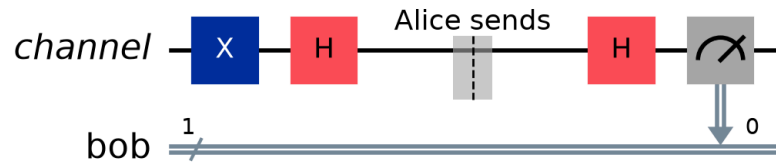
```
qc.ry(theta, psi)           # the state to send
qc.h(a); qc.cx(a, b)        # shared Bell pair
qc.cx(psi, a); qc.h(psi)    # Alice: reverse Bell circuit
qc.measure(psi, c[0]); qc.measure(a, c[1])
with qc.if_test((c[1], 1)): qc.x(b)    # Bob corrects
with qc.if_test((c[0], 1)): qc.z(b)
```

Verified in `teleportation.py`: $|0\rangle$, $|1\rangle$, $|+\rangle$ and $R_y(\pi/3)|0\rangle$ all arrive in 100% of shots.

BB84 — detecting an eavesdropper

Not encryption. It produces a shared random key **and tells you if anyone looked**.

$Z = \{|0\rangle, |1\rangle\}$ and $X = \{|+\rangle, |-\rangle\}$ are **conjugate** — certain in one, a coin flip in the other.



One qubit, changing hands. `channel` is Alice's before the barrier, Bob's after.

- `X` — Alice's **bit**: present 1, absent 0
- first `H` — her **basis**: present X , absent Z
- second `H` — **Bob's** basis, picked blind

The order is the security:

1. Alice sends the qubit
2. Bob measures **immediately**, guessing a basis
3. *Then* they announce bases — never bits
4. Discard the mismatches

Bases match $\rightarrow$ the two H s cancel, $H^2 = I \rightarrow$ Bob reads her bit exactly. Bases differ $\rightarrow$ one lone $H \rightarrow$ a coin flip.

BB84 — eight rounds, start to finish

Alice flips **two** coins (bit, basis). Bob flips **one** (basis), blind.

#	A bit	A base	sends	B base	B reads	match?	
0	1	X	$ -\rangle$	X	1	yes	key
1	0	Z	$ 0\rangle$	X	0	no	discard — right by luck
2	0	Z	$ 0\rangle$	X	1	no	discard — wrong
3	1	Z	$ 1\rangle$	Z	1	yes	key
4	0	X	$ +\rangle$	X	0	yes	key
5	0	Z	$ 0\rangle$	Z	0	yes	key
6	0	Z	$ 0\rangle$	Z	0	yes	key
7	0	Z	$ 0\rangle$	X	0	no	discard

They announce their **bases** publicly — never the bits — and drop the mismatches.

Alice keeps **1 1 0 0 0**, Bob keeps **1 1 0 0 0** : a secret neither of them sent.

Rounds 1 and 2 are the point — same choices, different outcomes. On a mismatch Bob still gets a bit, just a worthless one, and nobody knows which until the bases are announced.

BB84 — what actually becomes key

A real run of `bb84.py --rounds 10` :

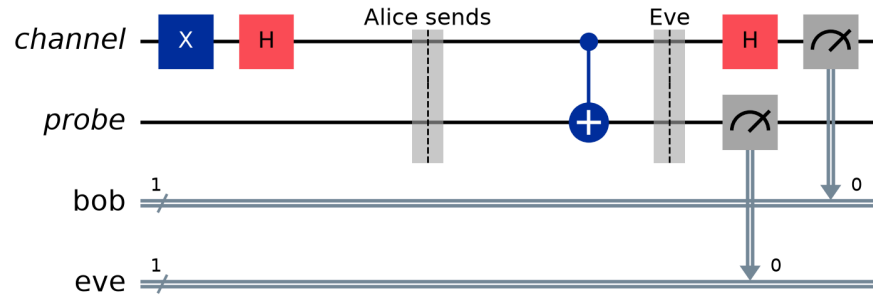
```
transmitted          10
bases agreed          7      <- sifting: ~50% on average
sacrificed to test    4      <- announced openly, so spent either way
left for the key       3
```

Two losses. Half the rounds die at sifting, because the bases disagreed. More are sacrificed to test for Eve — once announced they are public, so they are spent whatever they show. Roughly **a quarter** of transmitted qubits become key.

Each compared bit is an independent $\frac{3}{4}$ chance Eve slips past, so detection needs volume:

bits compared	4	50	126
Eve escapes undetected	32%	6×10^{-7}	2×10^{-16}

BB84 — what Eve costs



Eve cannot copy the qubit (**no-cloning**), so she must measure — and guess a basis.

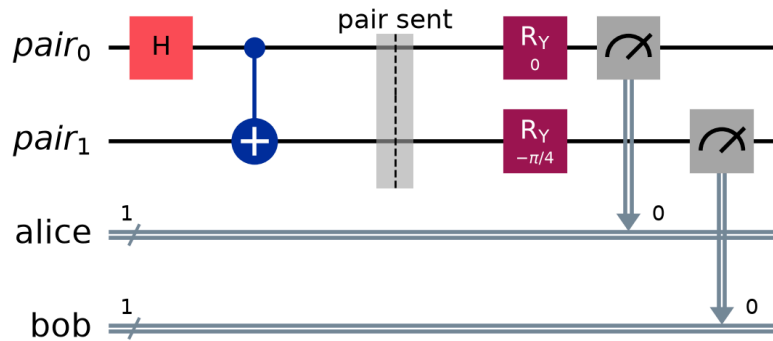
She is wrong half the time, and then Bob errs half of *those*:

$$\text{QBER} = \frac{1}{2} \times \frac{1}{2} = 25\%$$

She does **not** change the sifting rate — only the error rate on the bits that survive: $0\% \rightarrow 25\%$. Alice and Bob compare some openly, and **abort above 11%**.

Her gain: 75% of the key — useless, because the key is discarded.

E91 — the same job, via Bell's theorem



A source emits entangled pairs. Each side measures along a randomly chosen axis.

$$E(\theta_a, \theta_b) = \cos(\theta_a - \theta_b)$$

Combine four settings into the **CHSH** sum S . Any theory where the bits existed before measurement obeys $|S| \leq 2$. Quantum mechanics reaches $2\sqrt{2} \approx 2.83$.

Eve has to entangle herself with the pair to learn anything — and **monogamy of entanglement** means that halves every correlation, dropping S to 1.41.

Measured: $S = 2.78 \pm 0.07$ clean, 1.34 with Eve. The eavesdropper is visible as a *failed violation*.

Recap

	sends	adversary	needs entanglement
Superdense coding	2 bits via 1 qubit	no	yes
Teleportation	1 qubit via 2 bits	no	yes
BB84	a secret key	yes	no
E91	a secret key	yes	yes

All four run locally or on IBM hardware with `--ibm` .

github.com/alessandrogpz/quantum-computing-notes